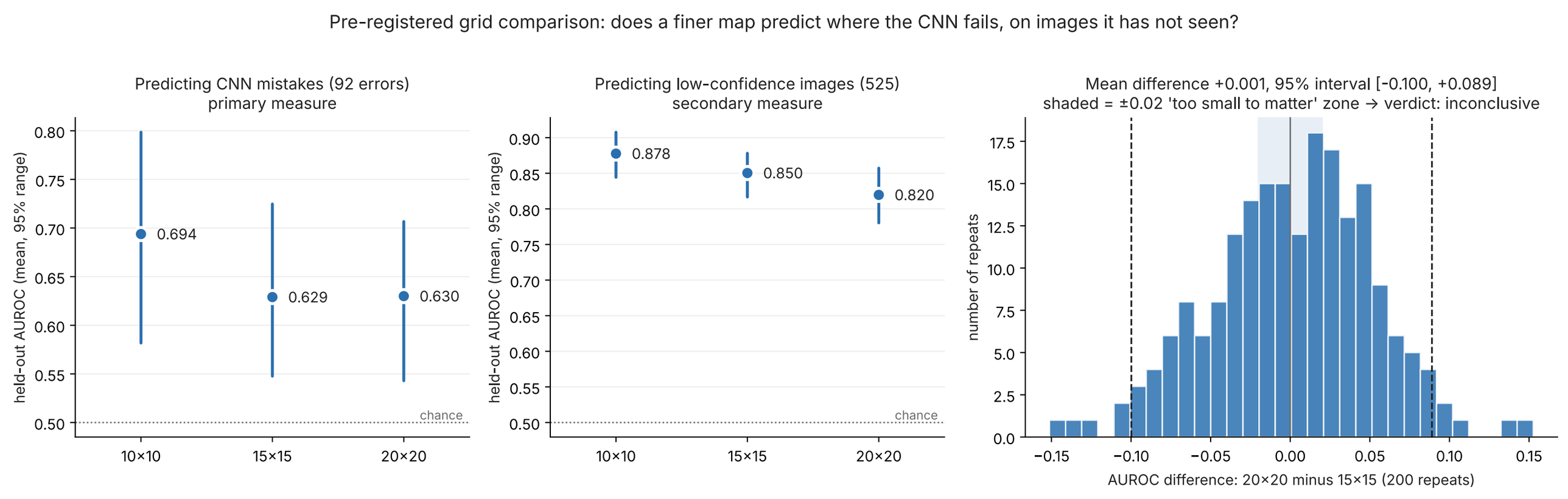

Pre-registered grid comparison: does a finer map predict where the CNN fails, on images it has not seen?
Predicting CNN mistakes (92 errors)
primary measure
held-out AUROC (mean, 95% range)
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.694
0.629
0.630
chance
10×10
15×15
20×20
Predicting low-confidence images (525)
secondary measure
held-out AUROC (mean, 95% range)
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.878
0.850
0.820
chance
10×10
15×15
20×20
Mean difference +0.001, 95% interval [-0.100, +0.089]
shaded = ±0.02 'too small to matter' zone → verdict: inconclusive
number of repeats
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
AUROC difference: 20×20 minus 15×15 (200 repeats)